Caspar Popova
Ph.D. Student in Programming Languages & Compilers | cjpopova.github.io | caspar@umd.edu

Education

University of Maryland, Ph.D in Computer Science — College Park, MD; 2024 - Present
Northwestern University, Bachelor of Science in Computer Science — Evanston, IL; 2019 - 2022
Relevant coursework: PL + Architecture (HLS), Compilers, Theorem proving (Roq, Lean), PL + Security, Systems for Machine Learning, Databases

Ongoing Research

Compiling OxCaml Correctly — September 2026 - Present
Testing the OxCaml compiler by defining specifications for layouts and developing generation strategies for moded programs.
→ Random Testing, Compiler Testing, OCaml

Does Rust Compile Volatile Correctly? — June 2026 - Present
Testing that rustc compiles volatile code for embedded devices correctly by combining binary instrumentation testing with random program generation.
→ Random Testing, Compiler Testing, Rust, C++, Program Instrumentation

Sized Types for Program Generation — May 2025 - Present
Testing recursion optimization in compilers by developing a program generator of terminating recursive programs using sized types.
→ Random Testing, Compiler Testing, OCaml, Racket, Python

A Learned Cost Model for PyTorch Kernel Fusion — October 2026 - Present
Replacing PyTorch's heuristic kernel fusion ranking strategy with a learned cost model using the program's memory behavior and compute characteristics.
→ PyTorch, Kernel Optimization, ML Compilers, Python

Research Experience

Recursion for HLS in PandA — Oct 2025 - Dec 2025
Extended the open source PandA HLS framework with support for recursion by implementing a recursion-to-iteration compiler pass.
→ Compilers, HLS, GCC, C++

PLT Group, Northwestern University — Evanston, IL; 2021-2022
Designed and implemented an SMT solver-based program synthesizer for higher-order functional programs to evaluate the usefulness of error-reporting mechanisms in gradually typed languages including Typed Racket.
→ Program Synthesis, Gradual Typing, Solver-Aided Programming, Racket

Arcana Compilers Group, Northwestern University — Evanston, IL; Apr - Jun 2022
Implemented a compiler transformation that lowers object abstractions and performs stack vs. heap allocation analysis as part of the MemOIR compiler, which provides an intermediate representation for objects to enable specialized transformations like dead field elimination.
→ Compilers, LLVM, C++

Work Experience

Sandia National Laboratories — Remote/CA; June 2026 - Aug 2026
Graduate Research Intern
Working with John Bender to test whether the Rust compiler compiles volatile correctly.

PLUM Lab, University of Maryland — College Park, MD; Aug 2024 - Present
Graduate Research Assistant
Working with David Van Horn and Leonidas Lampropoulos on compiler testing.

Epic Systems — Verona, WI; 2022 - 2024
Software Developer in Population Health Analytics

- Drove multi-year strategic direction of an analytics framework via client feedback and collaboration with other analytics teams.
- Built population health data models and visualizations within Microsoft SQL EDW in collaboration with multiple teams.
- Designed internal processes and utilities to improve best practices, development, testing completeness, and coordination with other teams.

→ Database Management, Data Models, Analytics, SQL, C#, TypeScript, Dashboard Design

Talks

- Does Rust Compile Volatile Correctly? — SSNL 2026 (SPLASH-ISSTA)
- Compiler Testing with Sized Types — NJPLS December 2025

Awards & Honors

- 2023 CSGrad4US Fellow (NSF-sponsored fellowship similar to the GRFP)

Service & Teaching

- Teaching assistant for Design and Implementation of Programming Languages, Organization of Programming Languages (OCaml, Rust, Racket) in Fall 2024 - Spring 2025.
- Mentoring new Ph.D. and M.S. students at University of Maryland.
- Student volunteer at POPL 2025.
- Mentoring new software developers at Epic Systems.